

# Exercises in Marine Ecological Genetics

## 08. Selection

- Window-based genome scans
- $F_{ST}$  outlier loci
- Candidate genes under selection
- Haplotype-based selection tests

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<https://github.com/mhelmkampf/meg26>



# Today's objectives

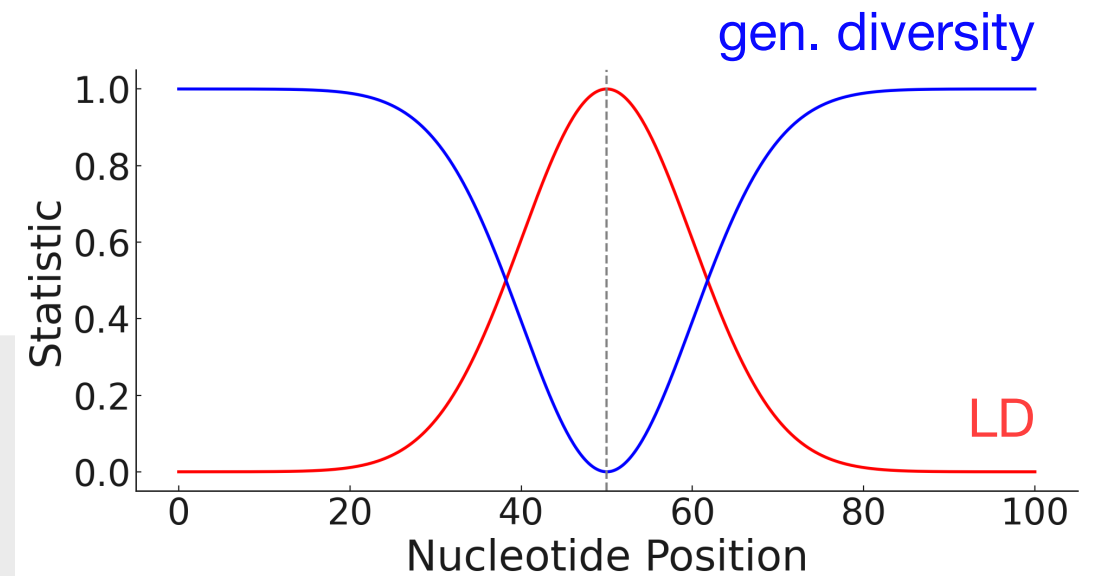
By the end of this session, you will be able to:

- Perform a **genome-wide scan for selection**, choosing suitable windowing parameters and calculating population differentiation statistics (e.g.,  $F_{ST}$ )
- Detect and identify  **$F_{ST}$  outlier loci** as potential targets of selection
- Link outlier loci to **candidate genes** by querying gene annotation files (GFF) and functional databases
- Evaluate the strengths and limitations of  $F_{ST}$ -based scans relative to **haplotype-based methods** for detecting selection

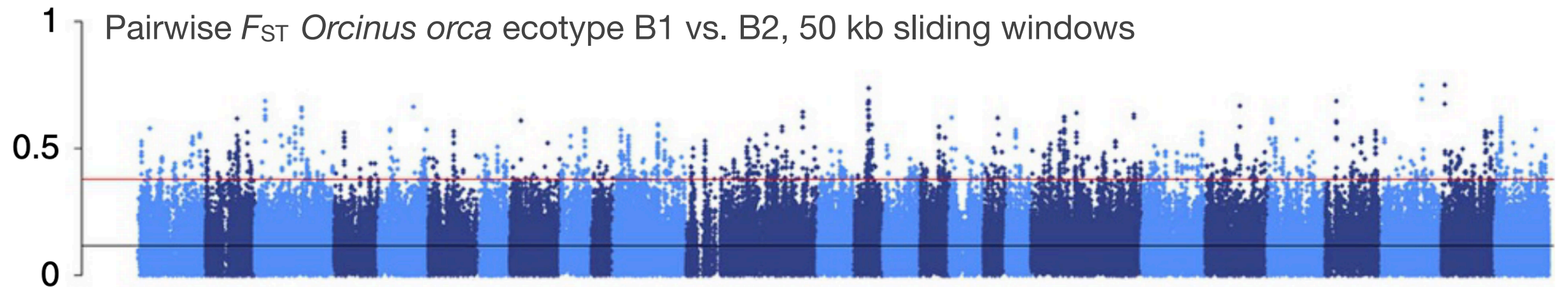
# Detecting selection

Positive selection influences genetic variation by:

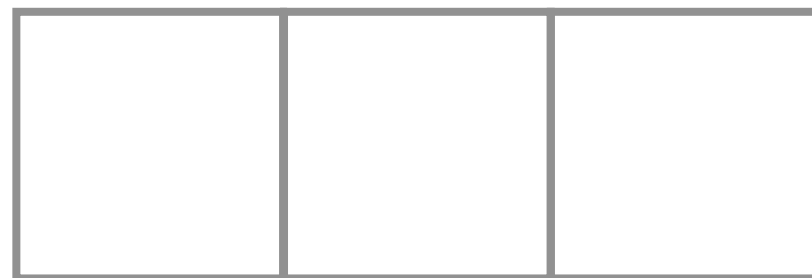
- Reducing genetic diversity (e.g.  $\pi$ )
- Increasing linkage disequilibrium
- Skewing allele frequencies / abundance of rare alleles (Tajima's D)
- Increasing population differentiation ( $F_{ST}$ )
- Extending homozygous haplotypes (iHS, EHH)



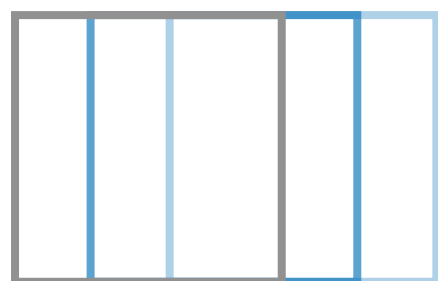
# Genome-wide scans using windows



Footo et al. 2016, *Nature Communications*



... Non-overlapping windows



... Overlapping / sliding windows



Window size



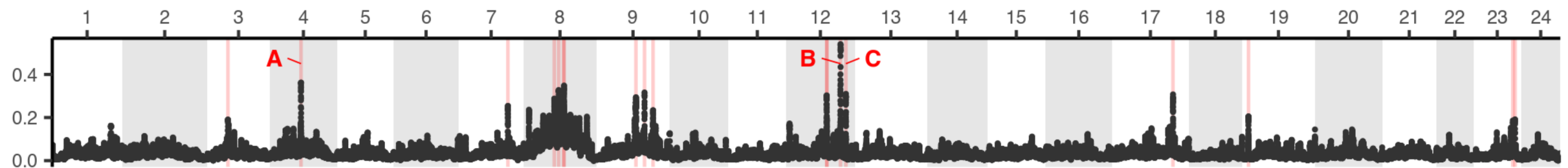
Increment / step size



# $F_{ST}$ scan across whole genome

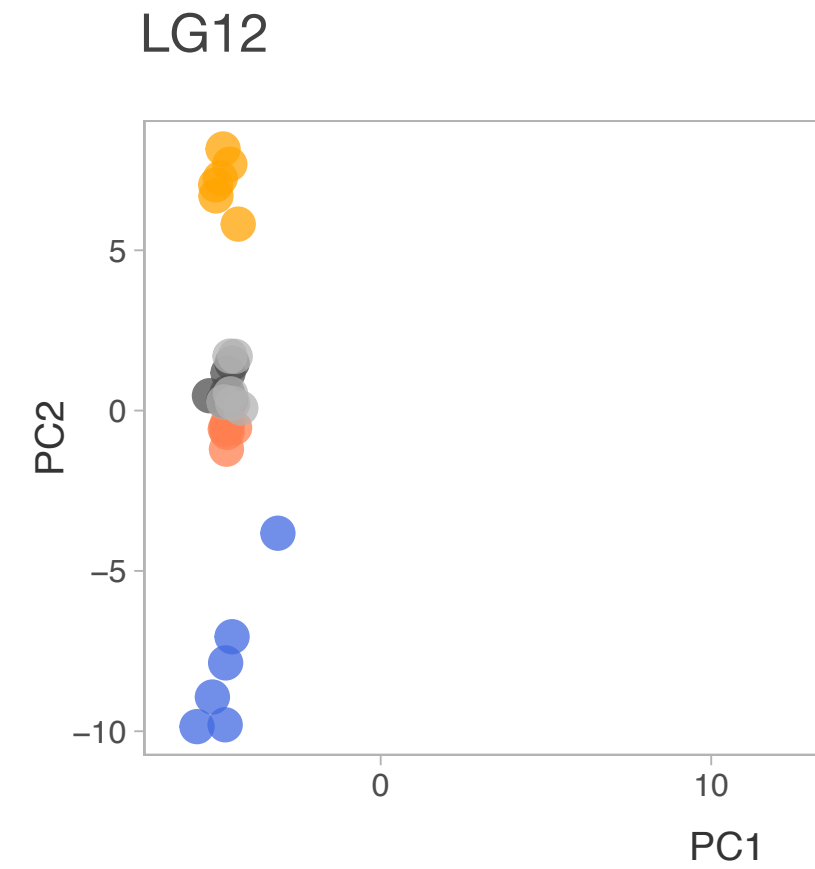
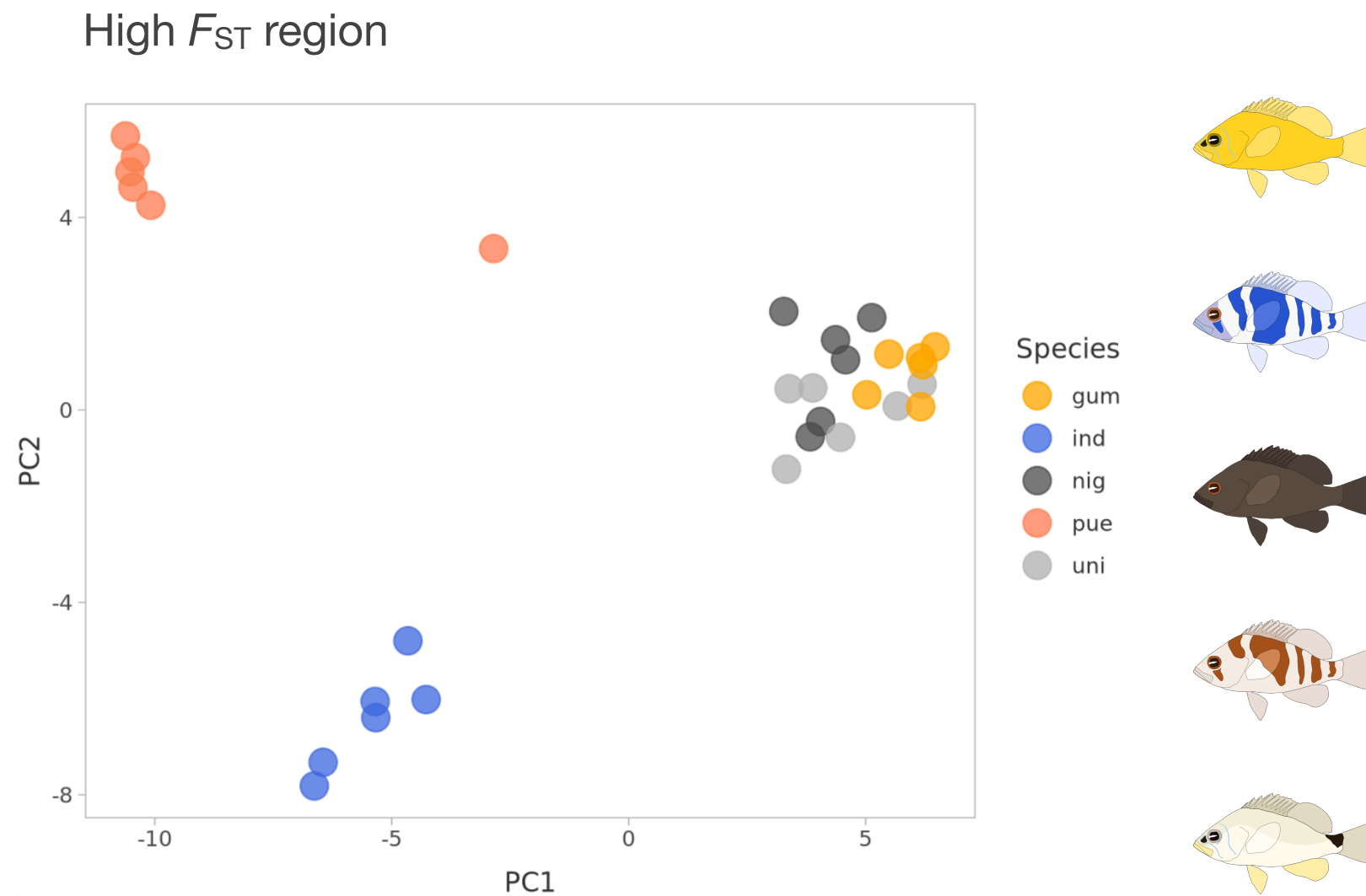
VCFTools, 50 kb sliding windows, joint  $F_{ST}$  of 14 Caribbean hamlet populations

Red = 99.8<sup>th</sup> percentile



Hench et al. 2022, PNAS

# PCA of $F_{ST}$ outlier region on LG12

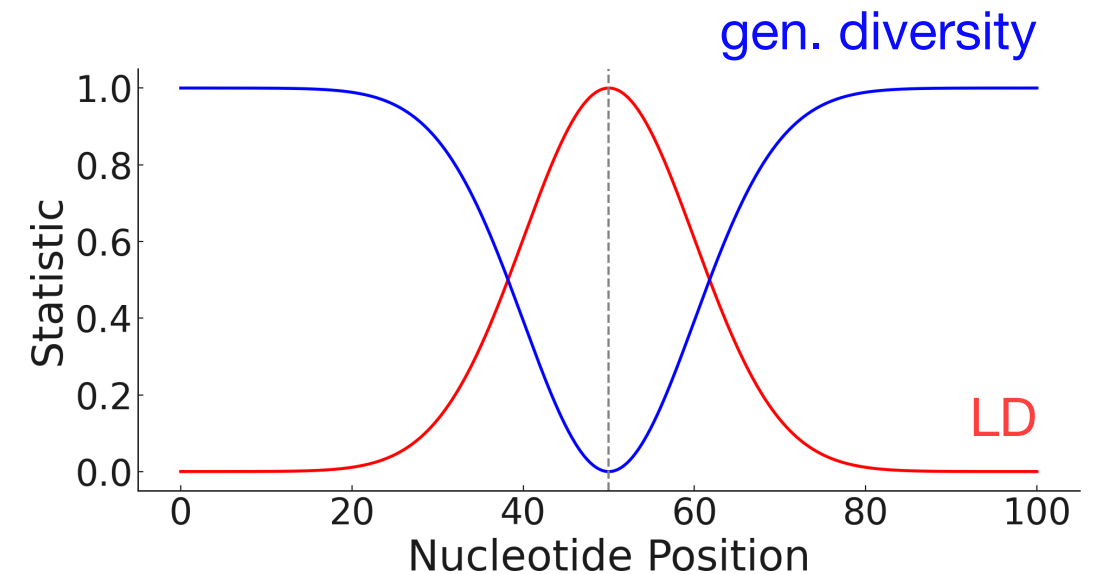


# General Feature Format (GFF)

```
##gff-version 3;Parentgene=;Parentgenename=  
Contig49      .          gene          406          8055          .          +  
      .          ID=HPv1G000000000001;...;Parentgenename=ZNF484  
Contig49      .          mRNA          406          8055          .          +  
      .          ID=HPv1M000000000001;...;Parentgenename=ZNF484  
Contig49      .          exon          406          700          .          +  
      .          ID=HPv1E000000000001;...;Parentgenename=ZNF484  
Contig49      .          exon          5740         5843          .          +  
      .          ID=HPv1E000000000002;...;Parentgenename=ZNF484  
Contig49      .          exon          5993         6110          .          +  
      .          ID=HPv1E000000000003;...;Parentgenename=ZNF484  
Contig49      .          exon          7271         8055          .          +  
      .          ID=HPv1E000000000004;...;Parentgenename=ZNF484  
...  
```

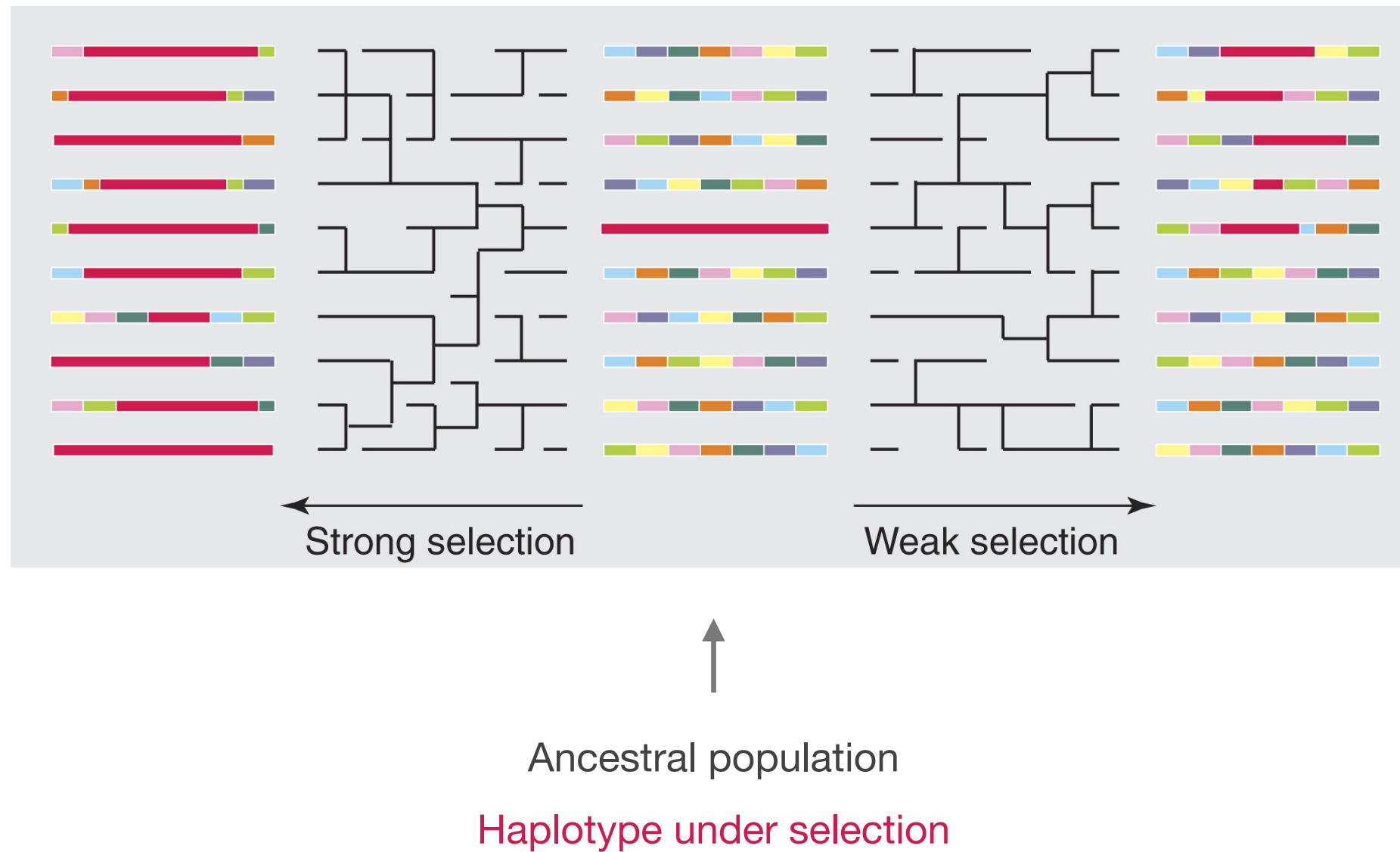
# Haplotype-based methods

- Positive selection increases the frequency of a beneficial mutation in the population
- Nearby loci “hitchhike” due to linkage disequilibrium, faster than recombination can break up the association (**selective sweep**)
- As the haplotype rises in frequency, more individuals carry the same extended haplotype around the selected site (**extended haplotype homozygosity**)
- Haplotype-based methods (e.g. iHS, XP-EHH) detect this pattern of long, high-frequency homozygous haplotypes as a signature of recent positive selection
- Longer homozygous haplotypes = stronger evidence for selection



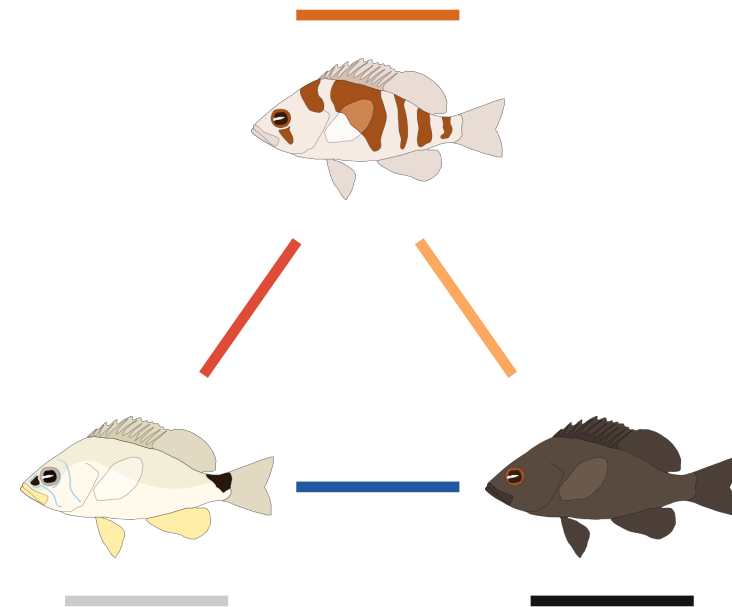
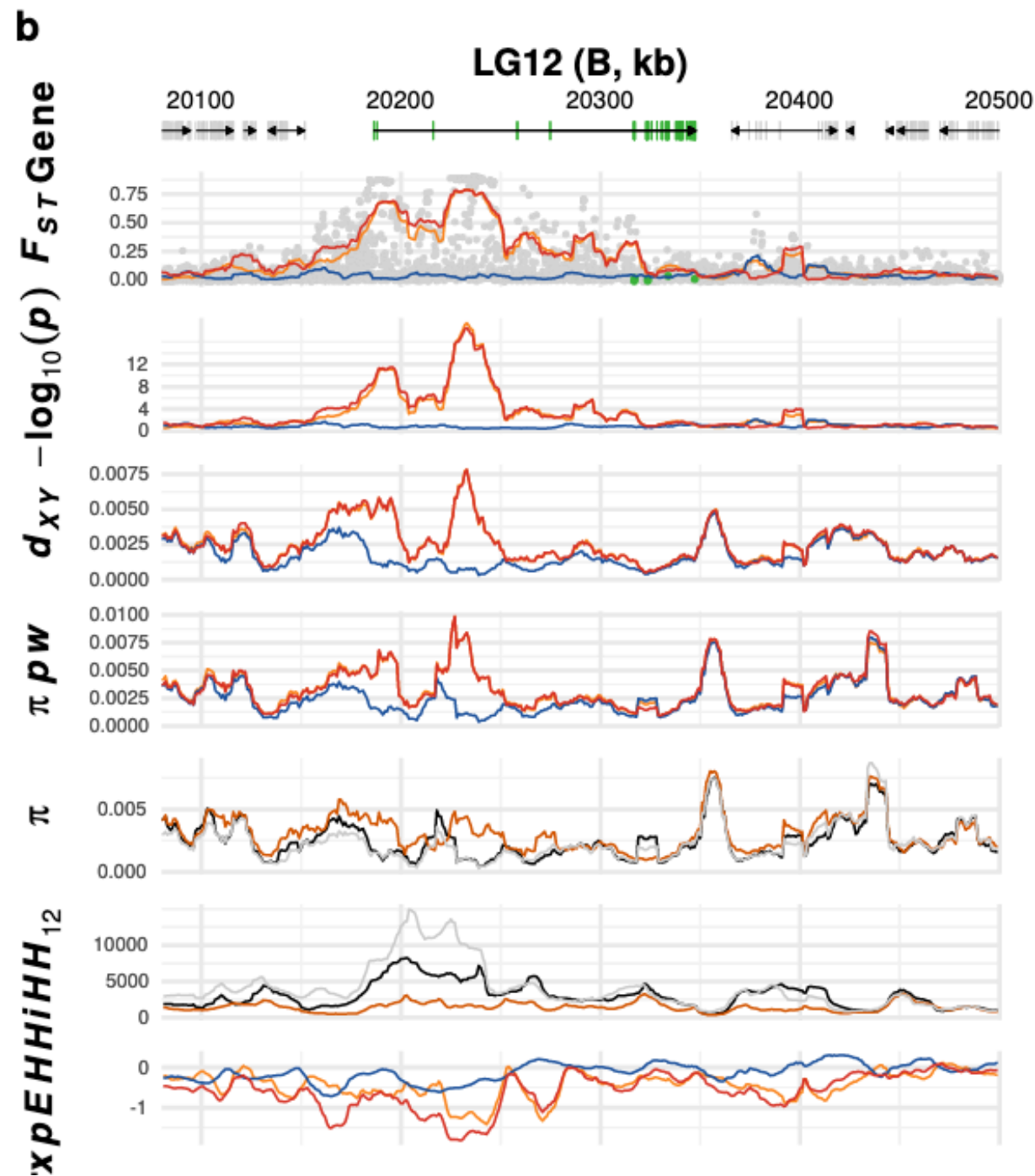


# Haplotype-based methods



Gibson 2007, *Current Biology*

# Genome-wide scans centered on *casz1* region



Hench et al. 2019, *Nat Ecol Evol*

# Phylogeny of *casz1* region

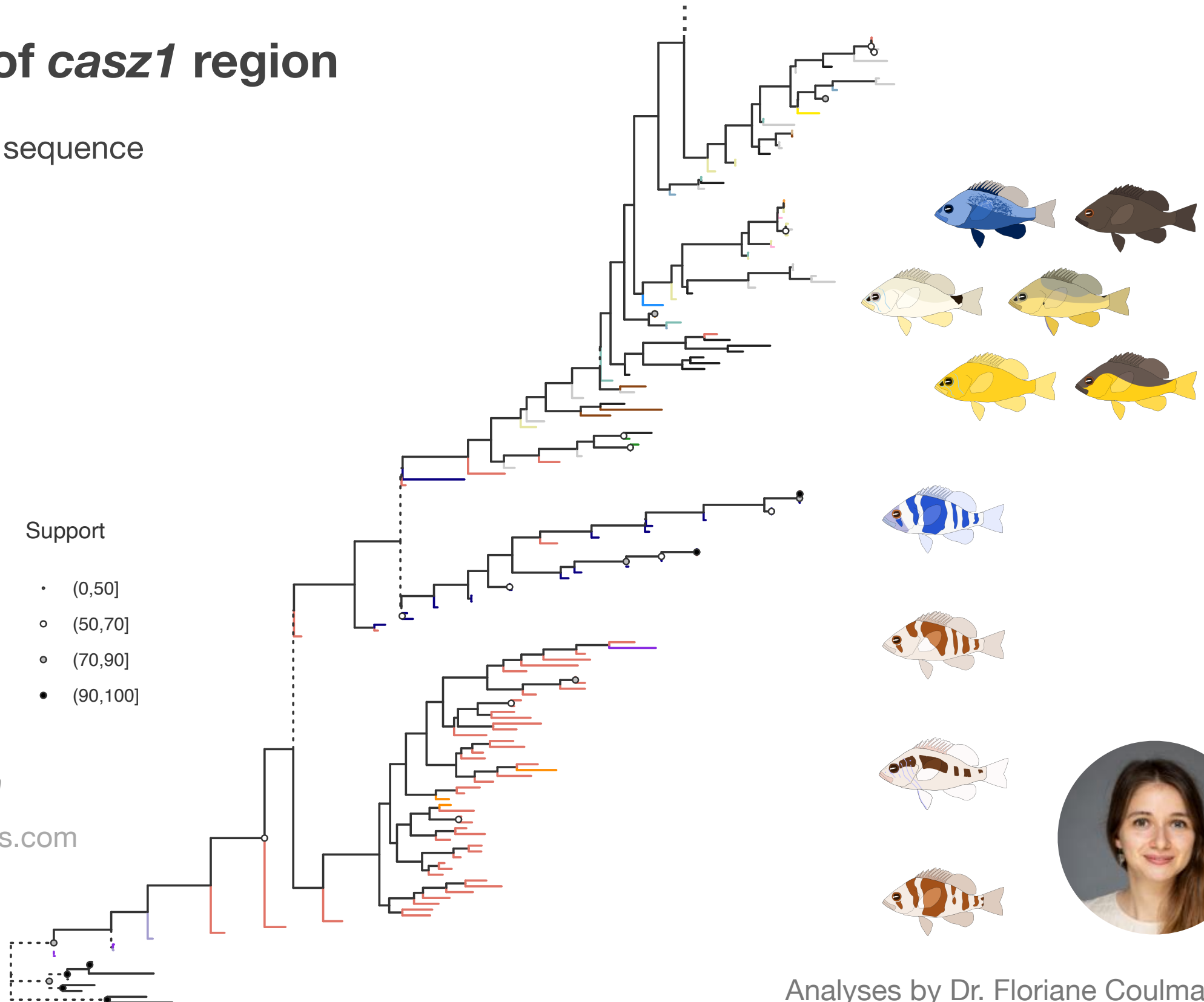
RAxML | ~ 200 kb sequence

Support

- (0,50]
- (50,70]
- (70,90]
- (90,100]

*Serranus tortugarum*

Image by whitecorals.com



Analyses by Dr. Floriane Coulmance

```
vcftools ... --weir-fst-pop --fst-window-step 5000 --fst-window-size 50000
```

```
scan_hh()          ### Scan for EHH (Extended Haplotype Homology), rehh package
```

```
manhattanplot()    ### Create Manhattan plot, rehh package
```

- Sliding-window scans help identify **regional genomic signals of selection** while reducing noise from individual markers
- Elevated  $F_{ST}$  values highlight candidate loci that **may be influenced by divergent selection**, but alternative explanations need to be evaluated
- **Functional gene annotation** transforms statistical signals into biologically meaningful hypotheses about adaptation
- $F_{ST}$ - and haplotype-based methods provide complementary insights because they detect different signatures and timescales of selection